

EX.NO: 8: INTER-THREAD COMMUNICATION

PROGRAM

```
class Railway {
    int seats = 2;

    synchronized void bookTicket(String name) {
        while (seats == 0) {
            try {
                System.out.println(name + "
waiting for a seat.");
                wait();
            } catch (InterruptedException e) {
                System.out.println(e);
            }
        }

        seats--;
        System.out.println(name + " booked a
ticket.");
        System.out.println("Available seats: "
+ seats);
    }

    synchronized void cancelTicket(String name)
{
        seats++;
        System.out.println(name + " cancelled a
ticket.");
        System.out.println("Available seats: "
+ seats);
        notifyAll();
    }
}

class Booking extends Thread {
    Railway railway;
    String name;

    Booking(Railway railway, String name) {
        this.railway = railway;
        this.name = name;
    }

    public void run() {
```

```

        railway.bookTicket(name);
    }
}

class Cancellation extends Thread {
    Railway railway;
    String name;

    Cancellation(Railway railway, String name) {
        this.railway = railway;
        this.name = name;
    }

    public void run() {
        try {
            Thread.sleep(1000);
        } catch (InterruptedException e) {
            System.out.println(e);
        }

        railway.cancelTicket(name);
    }
}

public class RailwayBooking {
    public static void main(String[] args) {
        Railway railway = new Railway();

        Booking b1 = new Booking(railway,
"Passenger 1");
        Booking b2 = new Booking(railway,
"Passenger 2");
        Booking b3 = new Booking(railway,
"Passenger 3");

        Cancellation c1 = new Cancellation(
railway, "Passenger 1");

        b1.start();
        b2.start();
        b3.start();
        c1.start();
    }
}

```

OUTPUT

```
Passenger 1 booked a ticket.  
Available seats: 1  
Passenger 2 booked a ticket.  
Available seats: 0  
Passenger 3 waiting for a seat.  
Passenger 1 cancelled a ticket.  
Available seats: 1  
Passenger 3 booked a ticket.  
Available seats: 0
```